

APPENDIX J

Optics

J.1. Optical sensors

Many attitude sensors are optical. Spacecraft optical sensors work in the infrared and visible bands. This appendix describes optical systems including lenses and telescopes. It also provides an introduction to geometric optics.

J.1.1 Optical nomenclature

Table J.1 gives optical concepts relevant to spacecraft sensing.

J.1.2 Telescope types

An imaging optical sensor consists of a telescope and an imaging chip. Fig. J.1 shows a hierarchy of telescope types. At the primary level, telescopes can be classified based on whether they employ reflective (mirrors) elements or refractive elements (lenses), or a combination of both. The greatest variety of telescopes is found among the reflectors. Fig. J.2 shows the optical geometry of the telescopes shown in Fig. J.1.

1. Refracting telescopes use a combination of an objective lens and an eyepiece lens. The objective lens defines the aperture of the telescope. The eyepiece lens causes the rays to converge. Refractors have relatively high spherical aberration and chromatic aberration, which is worse with short focal lengths. They also have light losses through the objective lens.
2. The Schmidt–Cassegrain has a corrector plate as the first optical element. This corrects for the spherical aberration of the primary mirror. It has light losses due to the obstruction.
3. The Maksutov–Cassegrain uses a meniscus corrector shell that is a section of a sphere. It has a spherical secondary mirror. The Maksutov has better resolution than the Schmidt. It suffers some light losses through the first optical element.
4. The Ritchey–Chretien is a Cassegrain reflector with two hyperbolic mirrors instead of a parabolic mirror. It can be free of coma and spherical aberration.
5. The Dali–Kirkham uses a concave elliptical primary mirror and a convex spherical secondary mirror. It does not correct for offaxis coma or field curvature.
6. The Schiefspiegler uses a tilted mirror to avoid the secondary mirror. This eliminates diffraction patterns but leads to an increase in coma and astigmatism.
7. The Gregorian employs a concave secondary mirror.

Table J.1 Nomenclature.

Term	Description
aperture stop	is a ring at the aperture that limits the size of the aperture. This is seen in cameras. The radius of the first lens serves as an aperture stop. The aperture stop controls the number of rays reaching the imaging plane but does not block all rays from a point on the imaged object. This is always behind the first lens.
achromat	a lens that reduces chromatic aberration. Often a doublet with lenses of different glasses.
apochromat	a lens that has less chromatic aberration than an achromat lens.
chief ray	is a ray that starts at the edge of an object and passes through the center of the aperture.
chromatic aberration	is the phenomenon of different wavelengths of light focusing at different points since the index of refraction is wavelength dependent.
coma	is an aberration in which parallel rays (for example from a star), at an angle to the optical axis of a lens focus at different points even though they would focus at the same point were the rays parallel to the optical axis. This means that stars on the edge of the field of view will be v-shaped with a comet-like tail.
field stop	is a ring at the imaging plane that determines the field of view of the instrument. The boundary of the CMOS chip acts as a field stop. The field stop obstructs certain rays completely.
meridional ray	is a ray that is confined to the yz -plane in an optical system in which z is along the optical axis and y is perpendicular to the axis.
paraxial ray	is a ray that makes a small angle to the axis system so that the angle of incidence $\theta \approx \sin \theta$.
sagittal ray	is a skew ray that enters the entrance pupil at $y = 0$.
skew ray	is a ray that starts in the yz -plane but enters the entrance pupil with a nonzero x coordinate.
vignetting	is the reduction of an image's brightness near the periphery. It can be caused by external objects, the physical dimensions of multiple lenses, and by the angle dependence of the image-sensor elements. The use of microlenses can reduce the pixel problem. Optical vignetting can be cured by reducing the aperture diameter.

8. Nasmyth reflectors use the third mirror to reflect light to the side. This adds flexibility. A prism can be used so that multiple targets are available. However, the prism adds to light losses.

Note that when the image is only to be viewed through an imaging chip no eyepiece is required. The imager is placed at the focal plane of the telescope. The simplest telescope would be a single objective lens and an imaging chip.

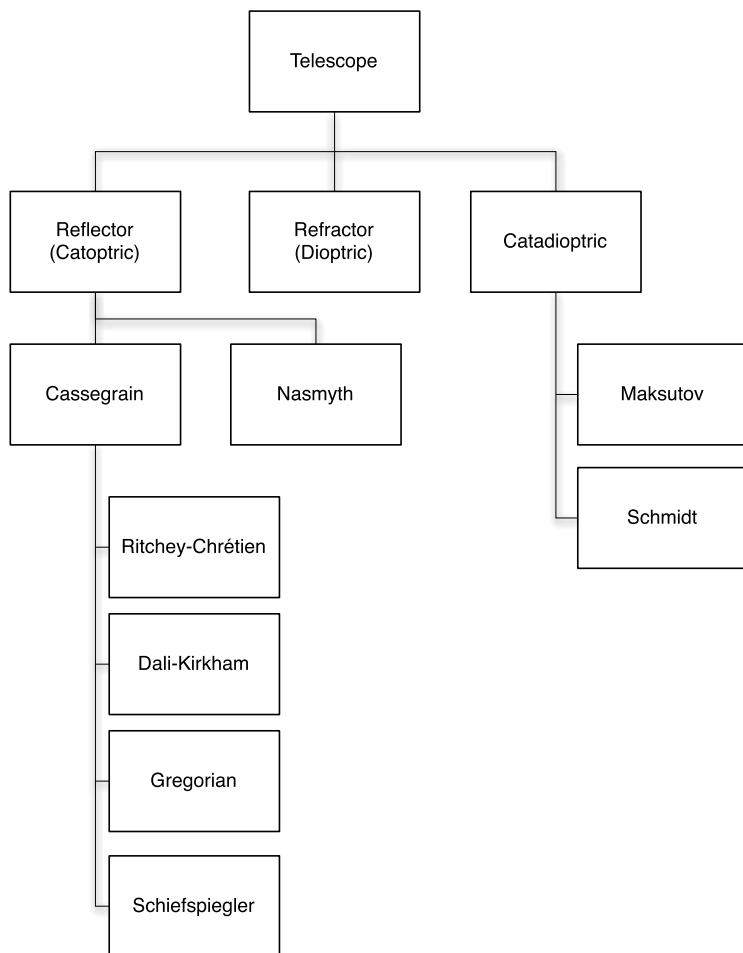


Figure J.1 Telescope taxonomy shows the variety of telescopes that could be used as part of a spacecraft optical system.

J.1.3 Geometry of imaging

The geometry of imaging is shown in Fig. J.3. This shows the relationship between the point on the image and the point on the focal plane.

In this diagram the subscript o is for object and i is for image. The blue lines are rays from the object point and all converge on the same point on the image plane. The position of the point r_o on the image plane is

$$|r_i| = r_o \left(\frac{f}{s_o - f} \right) \quad (\text{J.1})$$

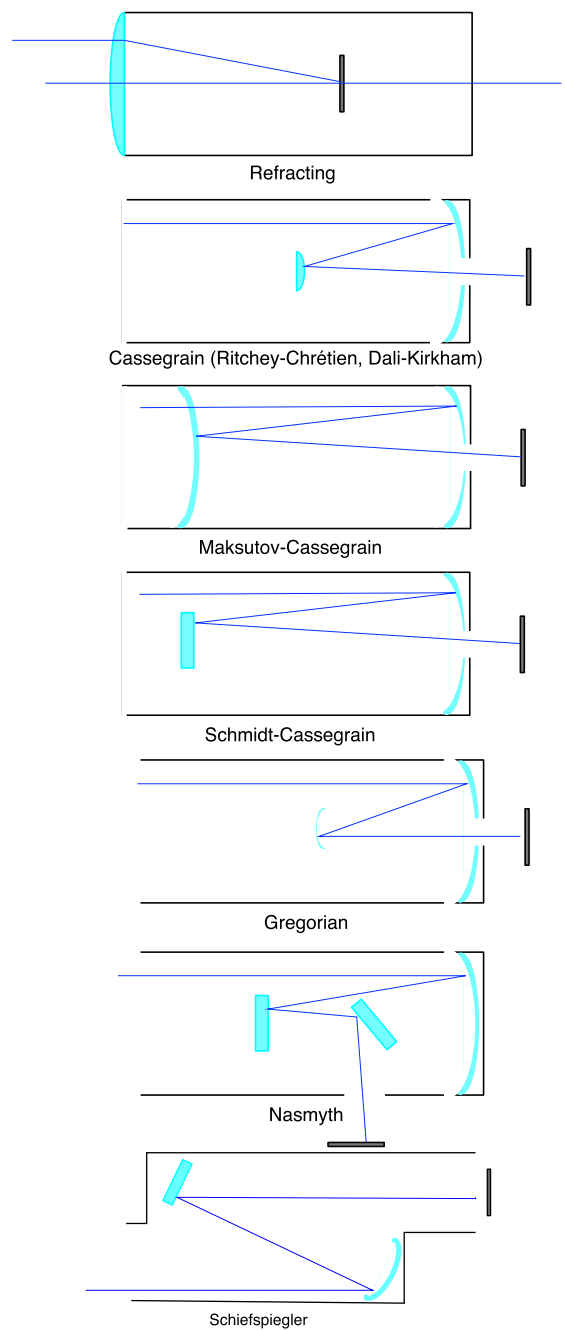


Figure J.2 Telescope optical geometry. The gray rectangle is the imaging chip.

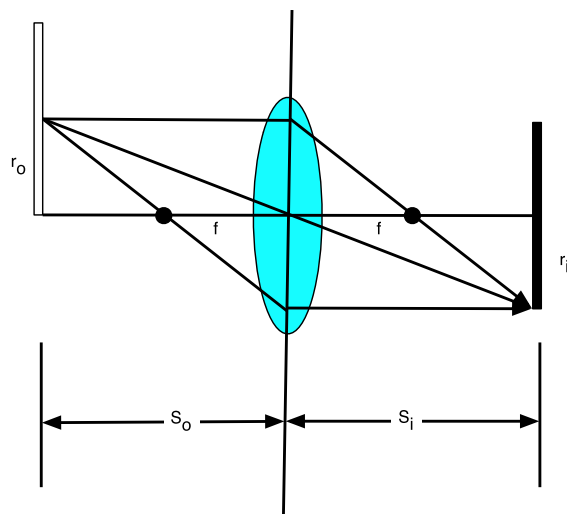


Figure J.3 Geometry of imaging for a simple lens.

where f is the focal length of the camera and s_o is the distance of the object from the camera lens. If the lens is circularly symmetric, then r_i is the distance of the point from the imaging axis. This formula assumes thin lens. The lens formula is

$$\frac{1}{f} = \frac{1}{s_o} + \frac{1}{s_i} \quad (\text{J.2})$$

This is identical for a mirror in which for a spherical mirror $f = -R/2$, where R is the mirror radius. The reflecting geometry is shown in Fig. J.4. If the target is many focal lengths away we find that

$$s_i = f \quad (\text{J.3})$$

Fig. J.4 shows the definition of the focal point for a mirror.

At a more fundamental level, optical-system performance is based on Snell's law of refraction [1] shown in Fig. J.5. We arrange lenses so that incoming rays from objects of interest arrive at the desired spot on the imaging chip.

$$\frac{\sin \theta_i}{\sin \theta_t} = \frac{n_2}{n_1} \quad (\text{J.4})$$

and Alhazen's law of reflection

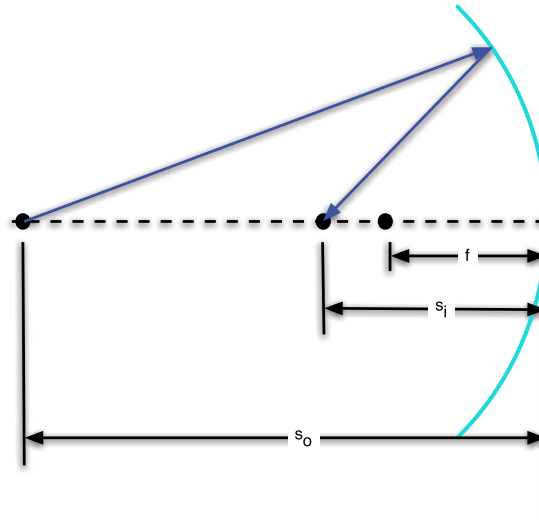


Figure J.4 Mirror optical geometry.

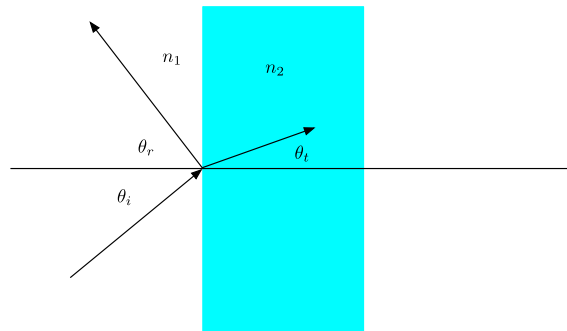


Figure J.5 Snell's law.

$$\theta_i = \theta_2 \quad (\text{J.5})$$

where n_k is the index of refraction. The index of refraction is a function of the wavelength of the incoming light. Different wavelengths of light will not focus onto the same spot when a single lens is employed. Different lenses of different refractive indices are used to compensate for this phenomena.

Lenses are either positive or negative [2], which defines the sign of their focal length. Either surface can be concave, convex, or flat, see Fig. J.6.

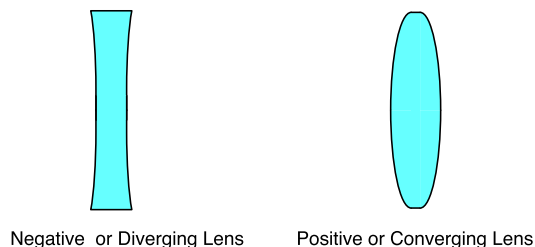


Figure J.6 A negative lens is on the left and a positive lens is on the right.

J.1.4 Telescope performance

J.1.4.1 Geometric

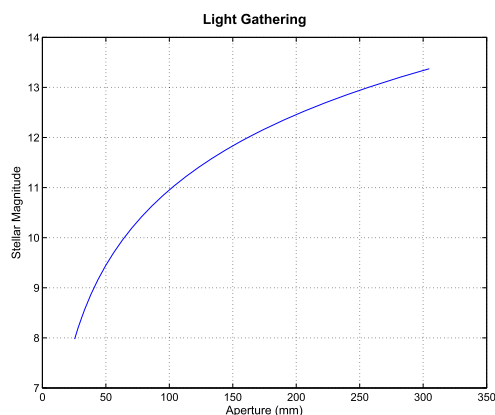
The angular resolution of a telescope is limited by the aperture. The Rayleigh criterion for the resolution limit is given by

$$\alpha_D = 1.22 \frac{\lambda}{D} \quad (\text{J.6})$$

where D is the aperture diameter and λ is the wavelength of the light.

```

1 lambda = linspace(400,700)*1e-9;
2 f = 100;% Focal length
3 a = 50;% Aperture
4 F = f/a;% FStop
5 d = AiryDisk( lambda, F );
6
7 Plot2D( lambda*1e9, d*1e6, '\lambda_\mu(m)', '
    Diameter_\mu(m)', 'Airy_Disk' );
8 PrintFig(1,1,1,'DiffractionLimitedResolution')
```



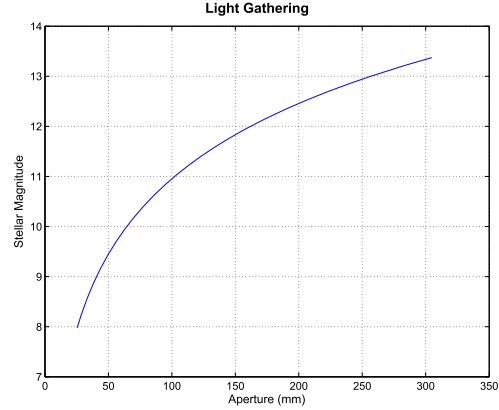
Example J.1: Diffraction-limited resolution.

This limit is based on the first zero-crossing of the Airy pattern for a point source (such as a star). This is shown in Example J.2. It shows the pattern with and without a central obstruction. Most reflecting telescopes have a central obstruction. It is possible to separate points that are closer than this value if the individual pixels are close enough together and the individual pixels have sufficient dynamic range. Such techniques are sometimes called superresolution techniques.

```

1 d      = 100;% Aperture
2 f      = 80;% Focal length
3 lambda = 650*1e-6;
4 eps    = 0.5;% Central obstruction
5 [i1, pixels] = AiryResolution( d, f, 3.5e-3,
                                lambda );
6 [i2, pixels] = AiryResolution( d, f, 3.5e-3,
                                lambda, eps );
7 Plot2D(pixels,[i1;i2], 'Pixels', 'Amplitude', 'Airy_
  Disk')
8 legend('Unobstructed', sprintf('%2.0f%% obstructed',
  eps*100))
9 PrintFig(1,1,1, 'AiryPattern')

```



Example J.2: Airy pattern for a point source.

The spatial resolution is

$$r_D = f \alpha_D \quad (\text{J.7})$$

The image size of an object of angular size θ is

$$w = f \theta \quad (\text{J.8})$$

If the sensor has p pixels per unit length then the angular resolution of each pixel is

$$\theta_p = f \theta / p \quad (\text{J.9})$$

For a star image the star will be a point and it will be desirable to defocus the point so that it is spread over several pixels. The light-gathering ability of a telescope is proportional to the square of the aperture

$$L \propto \pi \frac{D^2}{4} \quad (\text{J.10})$$

Losses include losses through lenses, the surfaces of mirrors and those due to obstructions (such as in a reflecting telescope) [3]. The best coatings on mirrors reduce losses to 2%. Losses in lenses are 1% per inch for inglass absorption. Light loss due to a central obstruction is

$$l_o = \frac{13a}{D} \quad (\text{J.11})$$

in percent where a is the diameter of the obstruction.

Both light gathering and resolution improve with larger apertures. However, larger apertures result in more mass and volume. The f-ratio or f-number of a telescope is

$$F = \frac{f}{D} \quad (\text{J.12})$$

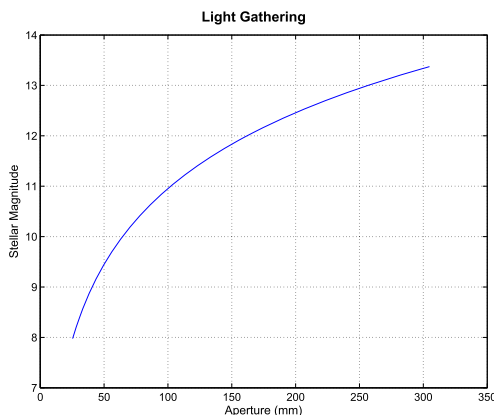
The greater the f-number the less light reaches a given area on the focal plane.

The stellar magnitude that can be seen by an ideal telescope is

$$m = 2.7 + 5 \log D \quad (\text{J.13})$$

where D is the aperture in mm. Example J.3 shows the light-gathering capability of a telescope.

```
1 LightGathering;
2 PrintFig(1,1,1, 'LightGathering');
```



Example J.3: Light-gathering capability of a telescope.

Example J.4 shows the visible stars as a function of telescope aperture using the Hipparcos star catalog. The sensor efficiency (the product of the fill factor and the sensor quantum efficiency) is 0.2. Three lenses are assumed, two for the objective and one for the field corrector each with a light-transfer efficiency of 0.99. A 160-mm aperture is sufficient to see the entire catalog.

J.1.4.2 Errors

The first-order monochromatic errors can be decomposed into the following five types: spherical aberration, coma, astigmatism, field curvature, and distortion.

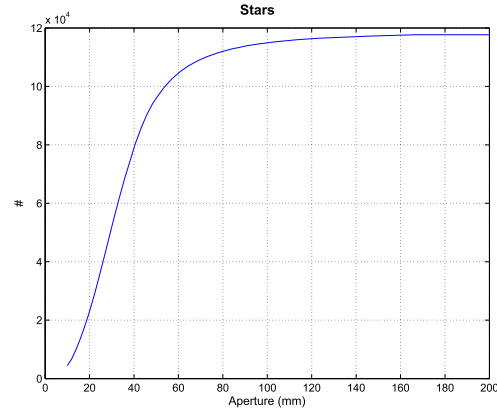
Spherical aberration is the difference in focal length between axial rays and rays along the periphery of aperture. These errors are proportional to the square of the aperture.

Coma is a geometric property in which offaxis point sources do not all come together at the same point. In an ideal telescope all of the light gathered by the aperture

```

1 nLenses = 3;
2 nSurfaces = 2*nLenses;
3 eff = 0.2*0.99^nSurfaces;
4 c = LoadCatalog('Hipparcos',12);
5 n = 100;
6 nStars = zeros(1,n);
7 d = linspace(10,200,n);
8 for k = 1:n
9     m = LightGathering(d(k), eff);
10    nStars(k) = length(find(c.vM<m));
11 end
12 Plot2D(d, nStars, 'Aperture(mm)', '#', 'Stars')
13 PrintFig(1,1,1,'NStars');

```



Example J.4: Visible stars for an ideal aperture.

from a source at infinity should appear at the same point. Schmidt, Maksutov, and other telescopes are designed to reduce coma.

Monochromatic or third-order astigmatism is a geometric effect even in symmetric optical systems. It is sometimes called third-order astigmatism. The larger the angle between the ray from the viewed object and the axis of the system the larger the astigmatism.

Field curvature means that the image really lies on the surface of a sphere, not a flat plate. This causes offaxis errors when an imaging chip is used.

Barrel distortion is an effect in which magnification decreases with distance from the optical axis. Pincushion distortion is the opposite.

There are also the chromatic aberrations decomposed into longitudinal and transverse. Additional errors are lens flare and diffraction.

Lens flare is light scattered in a telescope by internal reflection and scattering when a bright source, such as the Sun, is in the field-of-view. Lens flare often occurs in lenses with multiple elements such as zoom lenses. Lens flare can also occur when the bright object is not in the field-of-view in which case it appears as a haze that reduces contrast. Internal diffraction on the image sensor can cause small rainbows in the image.

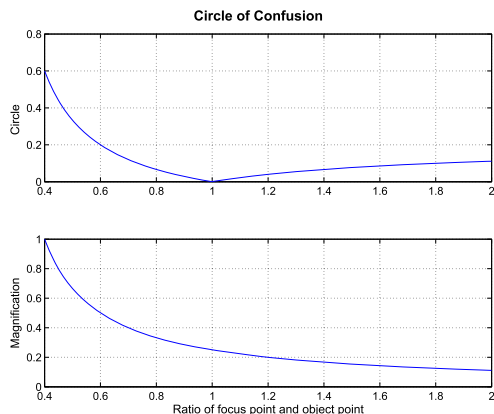
Diffraction spikes are an artifact caused by light diffracting around the support vanes of a secondary mirror in a reflecting telescope. Baffles and other objects can also cause diffraction errors. These patterns can cause errors in centroiding. These errors can be minimized by employing a centroiding algorithm that uses the point-spread function for a point source that will then ignore the spikes. This is also reduced if the image is defocused since only sharp objects have distinctive diffraction artifacts.

The circle of confusion is the spreading of an ideal point from the light from a lens, which is focusing a point source, not coming to a perfect point. This is also known as the blur circle. This is illustrated in Example J.5.

```

1 s2 = 10;% Object location
2 s1 = linspace(4,20);% Object point
3 a = 1;% Aperture
4 f = 2;% Focal length
5 CircleOfConfusion( s1, s2, f, a );
6 PrintFig(1,1,1,'CircleOfConfusion');

```



Example J.5: Circle of confusion.

J.1.5 Pinhole camera

The lowest-order approximation to the optical system is a pinhole camera. The pinhole camera is shown in Fig. J.7.

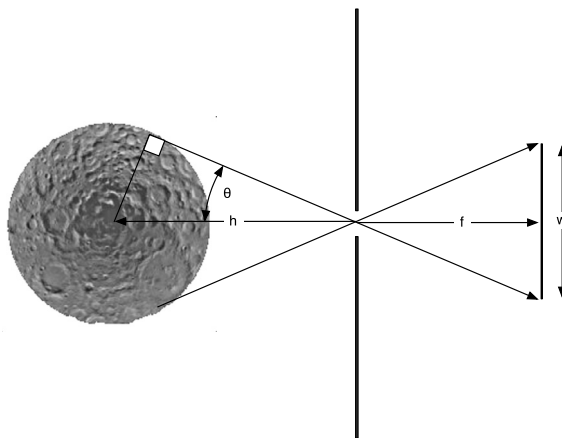


Figure J.7 Pinhole camera.

A point $P(X, Y, Z)$ is mapped to the imaging plane by the relationships

$$u = \frac{fX}{Z} \quad (\text{J.14})$$

$$v = \frac{fY}{Z} \quad (\text{J.15})$$

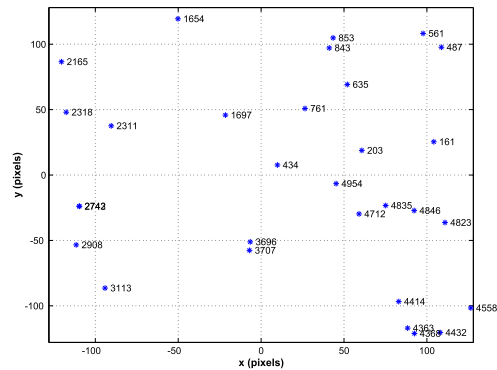
where u and v are coordinates in the focal plane, f is the focal length, and Z is the distance from the pinhole to the point along the axis normal to the focal plane. This assumes that the Z -axis of the coordinate frame X, Y, Z is aligned with the boresight of the camera.

The angle that is seen by the imaging chip is

$$\theta = \tan^{-1} \left(\frac{w}{2f} \right) \quad (\text{J.16})$$

where f is the focal length. The shorter the focal length, the larger the image. The pinhole camera does not have any depth of field, but that is unimportant for many far-field imaging problems. The field-of-view of a pinhole camera is limited only by the sensing element. An example of a pinhole image from the Hipparcos catalog (showing only stars with a visual magnitude >6) is given in Example J.6.

```
1 PinholeCamera
2 PrintFig(1,1,1,'PinholeCameraDemo');
```



Example J.6: Pinhole-camera star image.

J.2. Radiometry

J.2.1 Mathematical basis for position and attitude determination using a camera

The elements of a vision-based system are shown in Fig. J.8. The external sources of illumination for a satellite are nearby planets and the Sun. Since we are concerned about imaging we must consider the spectrum of the incoming radiation from each source. This plot gives the illumination spectrum. Radiation from the Moon and Earth is in the infrared part of the spectrum so will not be seen by cameras using conventional silicon imaging arrays. The Earth's black-body temperature is 295 K, the Moon's is 104 K, and the Sun's is 5760 K.

This also means that the illumination source for the spacecraft can be assumed to be a point, not a distributed source as radiation from a nearby planet would be modeled.

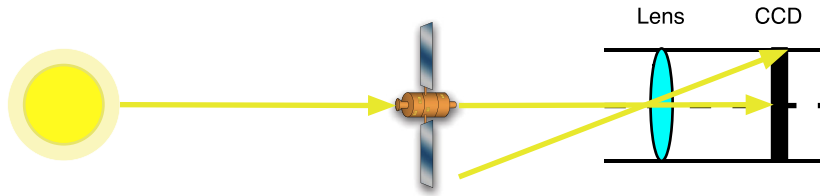
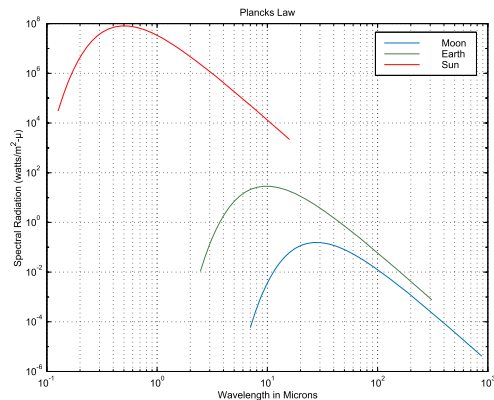


Figure J.8 Vision system.

```

1 tMoon = 104;
2 tEarth = 295;
3 tSun = 5760;
4
5 PlanckL ([tMoon; tEarth; tSun]);
6 legend('Moon', 'Earth', 'Sun');
7 PrintFig(1,1,1, 'PlanckEx')

```



Example J.7: Earth, Moon, and Sun spectra.

J.2.2 Basic radiometry

Fig. J.9 illustrates the basic radiometric concepts. The simplest illumination model is the Lambertian surface-reflection model

$$L = \rho I^T n \quad (\text{J.17})$$

where n is the normal to the surface, I is the vector to the illumination source, and $0 \leq \rho \leq 1$ is the surface's albedo.

According to the Lambertian model the Lambertian surface reflects light in a given direction d proportionally to the cosine of the angle between d and n . However, since the surface area seen from direction d is inversely proportionally to the angle between d and n , the angles cancel and d does not explicitly appear in the equation. In Fig. J.9 we have shown one source (the Sun); however, reflections from other surfaces will also be sources, thus complicating the illumination model. The energy received from the object is

$$E(p) = L(P) \frac{\pi}{4} \left(\frac{a}{f} \right)^2 \cos^4 \alpha \quad (\text{J.18})$$

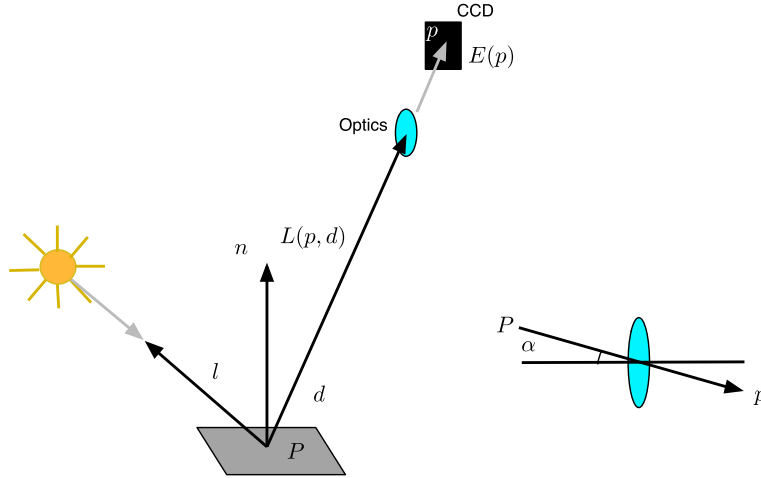


Figure J.9 Basic radiometry.

where a is the diameter of the aperture, f is the focal length, α is the angle between the incoming light and the optical axis, and f/a is the F-number of the optical system. For most optical systems α is small so the last factor is nearly 1. $E(p)$ is in units of W/m^2 while $L(p)$ is in units of $\text{W}/\text{m}^2 \text{ sr}$.

J.2.3 Radiosity

As mentioned above, each surface will be illuminated not just by the Sun, but also by radiation emitted from other surfaces. The Lambertian surface-reflection model assumes that the surfaces do not interact with each other and are not a source of illumination. A more general model is a radiosity model. Radiosity models are similar to models used for radiative thermal analysis of spacecraft.

The radiosity method generates a “photorealistic” model of the spacecraft that can be used as part of the vision system. There are two major parts to radiosity-based imaging:

1. Generation of form factors;
2. Solution of the radiosity equation.

The form factor between two planar surfaces is

$$F_{E_i \rightarrow E_j} = \frac{1}{A_i} \int_{A_i} \int_{A_j} \frac{\cos \theta_i \cos \theta_j}{\pi r^2} dA_i dA_j \quad (\text{J.19})$$

Note that this requires numerical integration over the finite area. In practice, if the surfaces are separated by more than five times their area, the integral can be replaced by a constant over the surfaces. The radiant exitance is then found from the radiosity

equation

$$M_0 = \begin{bmatrix} 1 - \rho_1 F_{11} & -\rho_1 F_{12} & \cdots & -\rho_1 F_{1n} \\ -\rho_2 F_{21} & 1 - \rho_2 F_{22} & \cdots & -\rho_2 F_{2n} \\ \cdots & \cdots & \cdots & \cdots \\ -\rho_n F_{n1} & -\rho_n F_{n2} & \cdots & 1 - \rho_n F_{nn} \end{bmatrix} \begin{bmatrix} M_1 \\ M_2 \\ \cdots \\ M_n \end{bmatrix} \quad (\text{J.20})$$

where

$$M = \pi L \quad (\text{J.21})$$

This matrix equation has one row and one column for each patch on the object. The column vector on the right is needed to compute the light reaching the camera. The vector on the left-hand side would include external sources, in this case the Sun. This equation is solved by a method called progressive refinement, which provides approximate solutions that improve with each iteration.

J.2.4 Radiometric sources

Radiometric sources for spacecraft optical include other spacecraft, the Sun, planets, and other solar-system bodies and stars. The stellar magnitude that can be seen by a telescope is shown in Example J.4.

Solar-system bodies will be illuminated by the Sun. Table J.2 lists the albedo and radiation reflected from each body at their mean distance from the Sun. The bond albedo is the ratio of the total radiation reflected to the incident radiation from the Sun. The geometric albedo is the fraction of the radiation relative to that from a flat Lambertian surface, which is an ideal reflector at all wavelengths.

The Sun's radiation in W/m^2 is

$$p = \frac{1367}{r^2} \quad (\text{J.22})$$

where r is distance from the Sun in astronomical units. 1367 is the flux of the Sun at the mean orbital radius of the Earth.

The albedo flux from any planet is

$$p = \frac{p_1 R^2}{r^2} \quad (\text{J.23})$$

where r is the distance to the satellite, R is the planet radius, and p_1 is the albedo flux at the surface of the planet.

Table J.2 Reflected light from planets and moons.

Planet	Geometric Albedo	Bond Albedo	Albedo Flux based on the Bond Albedos (W/m ²)
Mercury	0.138	0.056	510.873
Venus	0.84	0.720	1881.160
Earth	0.367	0.390	533.130
Mars	0.15	0.160	94.213
Jupiter		0.700	35.343
Saturn		0.750	11.272
Uranus		0.900	3.340
Neptune		0.820	1.240
Pluto	0.44–0.61	0.145	0.127
Europa		0.670	33.828
Triton		0.760	1.149
Earth's moon	0.113	0.120	164.040
Titan		0.210	3.156

J.2.5 Noise and performance factors

J.2.5.1 Fill factor

Fill factor is the percentage of the sensor that is active. In a CCD this is nearly 100%. In APS or CMOS sensors this can be considerably smaller due to the additional support circuits.

J.2.5.2 Illumination side

Chips can be back or front illuminated. Back illumination means the illumination is on the side opposite that of the circuit elements. This eliminates the blockage due to the traces increasing the fill factor to 100%. Most modern sensors use back-illuminated chips.

J.2.6 Dynamic range

The range of photon fluxes that can be accommodated without saturation. The dynamic range is determined by the size of the pixel and the number of electrons it can hold.

J.2.7 Blooming

Blooming is when charge from one detector overflows into another. In a CCD, if the cell saturates it will overflow into other cells. Antiblooming devices can be added to CCD chips at the expense of fill factor.

J.2.8 Quantum efficiency

Quantum efficiency is the percentage of photons hitting a detector that produce a hole–electron pair, that is an output. CCDs have quantum efficiencies of over 90% at some wavelengths making them well suited for low-light (or deep-sky) imaging. CMOS chips have quantum efficiencies that are around 30%.

J.2.8.1 Dark current

This is the constant output of a detector when it is not illuminated. The dark current will appear in every exposure. Since it is constant it can be measured and subtracted. This is best done on-orbit. CCD chips have very low dark current, as low as 1 electron/second at -30°C . CMOS cells are considerably higher. Part of this is due to the manufacturing of CMOS sensors in standard CMOS fabrication plants. CCD sensors that are used for imaging are fabricated on fabs specialized for imaging chips. As CMOS use increases dark current will improve. With all sensors dark current can be improved by cooling. For spacecraft applications and many terrestrial applications, thermoelectric coolers are used for imaging-chip cooling. These are solid-state devices and can be integrated into the other sensor electronics.

J.2.8.2 Fixed-pattern noise

Fixed-pattern noise is noise over the entire array that occurs in the same pattern on every exposure.

J.2.8.3 Readout noise

This is the noise added to a pixel output on reading it out.

J.2.8.4 Radiation hardness

Radiation hardness is the ability of the chip to withstand long-term and prompt radiation dosages. Prompt dosages can cause single-event upsets, in which a switch is set incorrectly, for example a readout switch. A greater problem is latchup that requires power to be cycled to restore the functioning of the chip. Long-term damage can increase the noise and reduce the quantum efficiency.

J.2.9 Imaging-chip theory

J.2.9.1 Dark current

The rate of dark-current creation is [4]

$$S = AT^{3/2}e^{-\frac{V_g q}{2kT}} \quad (\text{J.24})$$

where V_g is the gap voltage, T is the temperature in K, q is the electron charge (1.6×10^{-19} C), k is Boltzmann's constant (1.38×10^{-23} J/K), and A is 3.36×10^6 when S is

in volts. For a typical detector V_g is 1.36 V. To convert this to the number of electrons

$$e = \frac{St}{Cq} \quad (\text{J.25})$$

where C is the imaging-chip capacitance since

$$S = C \frac{dV}{dt} \quad (\text{J.26})$$

If the dark current is known it can be subtracted. However, this requires measurement of the current in the dark for each pixel. This is known as a thermal map. This equation shows it is advantageous to cool the chip, which can be done with a thermoelectric cooler. There are limits on how cold the chip can be cooled since the spectral sensitivity changes. Charges of thermal origin are created in the imaging elements and transfer elements.

J.2.9.2 Cosmic rays

A cosmic-ray impact will cause the generation of charge in a pixel. Cosmic rays are indistinguishable from stars. While not an issue with planetary imaging this can cause the sensor to misidentify stars. The simplest method is to take two images and compare them. The cosmic-ray image will only appear in one of the frames. The star identification software will generally also ignore “false stars”, particularly when the sensor is tracking known stars but the safest method is the two-frame method.

J.2.9.3 Thermal noise

Thermal noise is a function of the level of dark current. If N_t is the number of thermal charges created by the dark current then

$$\sigma_o = \sqrt{N_t} \quad (\text{J.27})$$

J.2.9.4 Transfer efficiency

Every time charge is transferred noise is created. The noise created is

$$\sigma_e = \sqrt{2\epsilon n N_s} \quad (\text{J.28})$$

where ϵ is the transfer efficiency, n is the number of transfers, and N_s is the charge transferred that includes thermal electrons. For an APS, CID, or CMOS sensor $n = 1$. For a CCD $n =$ the number of cells between the cell of interest and the readout.

J.2.9.5 Reset noise

At the output of a CCD register the floating readout diode is brought to a reference voltage before a packet is read out. This creates “reset noise”, which is

$$\sigma_r = \frac{1}{q} \sqrt{kTC} \quad (\text{J.29})$$

where C is the capacitance of the output diode. This noise can be eliminated by double-correlated sampling.

J.2.9.6 Photon noise

Photon noise is a result of photons hitting the chip and is

$$\sigma_s = \sqrt{QN_p} \quad (\text{J.30})$$

where Q is the quantum efficiency and N_p is the number of photons hitting the chip.

J.2.9.7 Quantization noise

Quantization noise is

$$\sigma_q = \frac{N}{2^n \sqrt{12}} \quad (\text{J.31})$$

where N is the maximum number of electrons per at the imaging-chip readout and n is the number of bits.

J.2.9.8 Total noise

The total noise is found by summing the squares of the noise sources

$$\sigma_t = \sqrt{\sigma_o^2 + \sigma_e^2 + \sigma_r^2 + \sigma_s^2 + \sigma_q^2} \quad (\text{J.32})$$

J.2.9.9 Blooming

When a photoelement is subject to strong illumination it can overflow and spill charges into adjacent photoelements if there is a direct connection between photoelements as there is in a CCD. CMOS, APS, and CID sensors do not have such a path and are immune to blooming. Another effect is that the charge cannot be emptied in one transfer leaving a “trail” of charges in that element. These leftover charges must be removed during successive readouts but it means that each readout will be contaminated. This is not to say that there is no crosstalk between sensing elements. All APS and CMOS chips specify crosstalk as a function of element separation.

CCDs can employ antiblooming devices to drain the overflow. Some of the excess charge will still remain. In addition, these devices take up space on the chip reducing its overall light-gathering capability. Typically, such devices reduce the sensitivity by 10%.

J.2.9.10 Linearity

The response curve of a photoelement is not entirely linear. At high levels the elements saturate. At low levels, the response is not proportional to the input. In most photoelements the nonlinearity at low signal levels is negligible. As an example the Cypress STAR 1000 sensor has a full-well capacity of 135 000 electrons but is linear to within 1% only with 95 000 electrons.

J.2.9.11 Amplifier noise

CMOS image sensors suffer from higher noise than CCDs due to the additional pixel and column amplifier transistor noise.

J.2.10 Data reduction

The following steps are taken to reduce data for an imaging chip [5]. Not all are applicable to CMOS chips. Several of the steps require calibration with either uniform sources or no illumination.

1. Bias subtraction – this is the removal of a fixed voltage to all elements. This is done by the subtraction of a bias frame.
2. Dark-current correction – this is the removal of the dark current. This is done by measuring the elements at the same temperature at which the exposure is taken and subtracting the dark frame. Hole-accumulation diodes (HAD) patented by Sony, can reduce the dark current at the source.
3. Flat-field correction – several images are taken under uniform illumination. These frames are averaged to produce the flat-field frame in which the frame contains the variations due to chip imperfections. The corrected frames (from the two steps above) are divided by this frame.
4. Cosmic-ray correction – this is done by taking several images of the same scene and removing artifacts that do not appear in all of the images.
5. Saturation trail – if a pixel is saturated the extra electrons will spill along the row. Any narrow feature along a row should be removed by postprocessing. This is a problem with CCD chips and does not appear in CMOS chips.
6. Diffraction features – support structure, baffles, and other objects can cause diffraction features. These patterns can cause errors in centroiding. These errors can be minimized by employing a centroiding algorithm that uses the point-spread function for a point source that will then ignore the spikes seen in the picture. This is also reduced if the image is defocused since only sharp objects have distinctive diffraction artifacts.

7. Bad column - in a CCD if a column is bad it will appear black and needs to be removed from the image. Any objects that intersect the bad column will be difficult to centroid. If a bad column is detected then the telescope should be pointed so that the objects of interest (for example an asteroid) do not appear in that column. This does not happen in CMOS chips.
8. Trap - in a CCD a damaged pixel may prevent reading of any pixels below that pixel. This will create artifacts in the image that must be avoided. This does not happen in CMOS chips.
9. Dust - dust on the lens will appear as constant dark areas. The flat-field correction will partially correct for these artifacts.
10. Photometric correction - Using objects of known brightness convert electrons produced by the detectors to flux.

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